Rota — Literature, Data & Technology Review

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Project at a glance

Rota detects the moment a university student drifts away from the reason they opened a video or social media app, and offers a gentle way back. When the app opens, the student types a short intention ("lentil soup recipe"). Rota compares each video they then watch with that intention and, when the session drifts, shows a reminder with two choices: return to the route, or continue on purpose.

Problem	Compulsive, unintended scrolling on YouTube and short-video feeds
Target group	University students in Türkiye (18+)
Core idea	Measure drift from a stated intention, not total screen time
Data	Open app-usage logs (LSApp) now; our own pilot with intention logs and watch history next
Approach	Pre-trained sentence embeddings (deep learning) + gradient-boosted classifier (machine learning)

Part 1 — Literature Review

1.1 Introduction

Problem. We address compulsive, unintended use of video and social media feeds: a student opens YouTube or Instagram for one purpose and is carried by recommendations into a long session they did not plan. We call this *intention drift*. It sits within problematic smartphone use (PSU) and problematic social media use, but it focuses on one moment that existing tools miss: the gap between why a session started and where it went.

Target group. University students in Türkiye, aged 18 and over. They are heavy users of video platforms, they manage their own time with little outside structure, and as adults they can give their own informed consent.

Why it matters. Online video is close to universal in Türkiye: in 2025, 90.9% of people aged 16–74 used the internet, 72.9% used YouTube and 68.1% used Instagram (TÜİK, 2025). Across 41 studies of young people, about one in four showed problematic smartphone use, and PSU was linked to higher odds of depression (OR 3.17), anxiety (OR 3.05) and poor sleep (OR 2.60) (Sohn et al., 2019). For students this reaches sleep, attention, coursework and relationships.

Research questions.

1. How can intention drift be detected from data a student can share without exposing private content?
2. Does drift from a stated intention identify problematic sessions better than a simple time limit?
3. Which kind of reminder brings students back to their intention without becoming another source of compulsive use?

Why review the literature first. The field already offers validated scales, known risk factors and tested interventions. It also has known weaknesses, such as the poor fit between self-reports and logged behaviour. Reviewing it first lets us reuse what works, avoid measures that do not, and show that our gap is real.

1.2 Thematic review

We organise the literature in six themes, moving from what the problem is, to how it is measured, to what has been tried.

Theme A — Definitions and diagnostic criteria

The only digital behaviour with formal diagnostic status is gaming disorder. In ICD-11, WHO defines it by impaired control, increasing priority over other activities, and continuation despite harm, normally evident for at least 12 months (World Health Organization [WHO], n.d.). Smartphone and social media "addiction" have no diagnosis. Andreassen et al. (2016) surveyed 23,533 adults and found that addictive social media use and addictive gaming had different profiles (social media more linked to women, gaming to men). They argued that a single "internet addiction" construct is not warranted. For our project this means we describe *problematic use* and its mechanisms rather than diagnosing anyone.

Theme B — Measurement tools and scales

Kwon et al. (2013) built the 10-item Smartphone Addiction Scale–Short Version (SAS-SV) on 540 Korean adolescents (Cronbach's $\alpha = 0.91$; cut-offs 31 for boys, 33 for girls). Noyan et al. (2015) validated the Turkish SAS-SV with 367 university students ($\alpha = 0.87$; test–retest 0.93). This is the scale we will use for our own participants. However, Ellis et al.

(2019) compared common smartphone scales with logged Apple Screen Time data and found weak correlations. Measures that framed use as *habitual* predicted behaviour better than those framed as *addiction*. Self-report alone is therefore not enough, which is why Rota relies on logged behaviour.

Theme C — Consequences and risk factors

Sohn et al. (2019) pooled 41 studies with 41,871 young people. The median prevalence of PSU was 23.3%, and PSU was associated with depression, anxiety, perceived stress and poor sleep. Most studies were cross-sectional, so cause and effect remain open. Chiossi et al. (2023) ran an experiment with 60 participants. Only the TikTok condition, with its rapid switching between short videos, impaired prospective memory: the ability to remember and carry out a planned intention. This is direct evidence that short-form feeds erode exactly what Rota tries to protect.

Theme D — Intentional versus habitual use, and the role of design

Lukoff et al. (2023) surveyed 606 US YouTube users and ran a 3-week field experiment with 46 of them. They compared a recommendation-first version of YouTube (*Explore*), a search-first version without recommendations or autoplay (*Focus*), and a version that let users switch between the two. Focus beat Explore on sense of agency, satisfaction and goal alignment, although it did not reduce time spent. Noe et al. (2019) logged taps, scrolls and typing from 64 participants for 8 weeks. Interaction with social apps, and Snapchat in particular, was associated with higher addiction scores. Together these studies show that the *way* an app is used, and the design that shapes it, matter more than total time.

Theme E — Prevention and intervention approaches

The commercial app *one sec* inserts a short pause before a chosen app opens. Its developers report a 57% reduction in use of the target app (one sec, n.d.); we treat this as a developer-reported figure rather than independent evidence. Monge Roffarello and De Russis (2019) reviewed 42 digital wellbeing apps and 1,128 user reviews, then ran a 3-week field study with 38 participants. Users found the apps useful, but timers and dashboards did not help them build new habits and were seen as not restrictive enough. Read together with SwitchTube, the lesson for Rota is that support built into the moment of use works, while passive statistics do not.

Theme F — Data-driven detection

Noe et al. (2019) used interface events logged on the phone to find indicators of addiction without relying on questionnaires. Ellis et al. (2019) showed why this matters: logged behaviour and self-report scales often disagree. Neither study, however, asked users what they *intended* to do, so neither could measure drift from that intention.

1.3 Summary and synthesis

Study	Aim	Population and size	Method	Key finding	Main limitation
Kwon et al. (2013)	Short smartphone addiction scale	540 adolescents, Korea	Scale development	10-item SAS-SV, α = 0.91	One region, non-clinical
Noyan et al. (2015)	Turkish SAS-SV	367 university students, Türkiye	Validation study	α = 0.87, test–retest 0.93	One university
Andreassen et al. (2016)	Link to psychiatric symptoms	23,533 adults, Norway	Cross-sectional survey	Social media and gaming addiction differ	Self-report, cross-sectional
Ellis et al. (2019)	Do scales predict behaviour?	Smartphone users with Screen Time logs	Scales vs logged use	Weak correlations	iPhone users only
Noe et al. (2019)	Indicators of addiction	64 participants, 8 weeks	Interface event logging	Social apps, Snapchat linked to addiction	Small sample
Monge Roffarello & De Russis (2019)	Do wellbeing apps work?	42 apps, 1,128 reviews, 38 users	Review + 3-week field study	Apps do not build new habits	Short study
Sohn et al. (2019)	Prevalence and outcomes of PSU	41 studies, 41,871 young people	Meta-analysis	PSU ≈ 23%; linked to depression, anxiety, sleep	Mostly cross-sectional
Chiossi et al. (2023)	Short videos and intentions	60 participants	Lab experiment	TikTok-style switching impairs prospective memory	Lab setting, short exposure
Lukoff et al. (2023)	Goal-first YouTube design	606 surveyed, 46 in field study, US	3-week field experiment, 3 app versions	Search-first Focus mode beat recommendation-first Explore on agency and goal alignment	Proof of concept; did not reduce time spent

Where they agree. Problematic use is common among young people and linked to poorer mental health and sleep (Sohn et al., 2019; Andreassen et al., 2016). Social and recommendation-driven apps are where the problem concentrates (Noe et al., 2019; Lukoff et al., 2023). Design that keeps the user's goal in view improves their sense of agency (Lukoff et al., 2023).

Where they disagree. Scale-based studies treat "addiction" as a trait to score, while log-based studies find that scores barely predict behaviour (Ellis et al., 2019). Wellbeing apps are liked but do not change habits (Monge Roffarello & De Russis, 2019), whereas a goal-first interface does improve agency and goal alignment (Lukoff et al., 2023). The difference seems to be where the support sits: inside the session it works; on a dashboard after the fact it does not.

Common limitations. Most studies are cross-sectional, rely on self-report, or use small samples from one country. Few involve Turkish students, and none uses a stated intention as a baseline.

1.4 Research gap and conclusion

Key takeaways. Problematic use affects about one in four young people. Time totals are a poor proxy for harm, and self-reports are a poor proxy for behaviour. The harmful pattern is habitual, recommendation-driven use that pulls people away from what they meant to do. Support built into the moment of use works better than passive tracking.

The gap. No study we found asks users for their intention at the start of a session and then measures, content by content, how far the session drifts from it. Existing tools count minutes, block apps or pause every opening. None can tell a purposeful hour from an hour lost to drift, and none distinguishes "never reached the goal" from "reached it and then drifted".

Our contribution. Rota makes intention drift measurable. It turns a stated intention into a reference point, uses language models to compare it with each video watched, and intervenes only when drift happens. For research, this provides a new behavioural measure that does not depend on self-reported addiction scores. In practice, it offers students a respectful nudge instead of a ban, and a route to Yeşilay's YEDAM services when patterns look serious.

Part 2 — Data Research

2.1 Introduction

Our data research has two goals. First, test on real logs whether the pattern Rota targets exists: sessions that start with a task and wander into a feed. Second, design the dataset we need to train and evaluate Rota itself.

The data should help answer three questions:

1. How common are feed sessions, and are they longer than other sessions?

2. When a session starts in a task app, how often does it end up in a feed, and what does that do to its length?
3. What would a labelled intention-drift dataset need to contain?

Exploring the data first matters because a model is only as good as its inputs. App logs carry duplicates, empty sessions and time-zone issues, and our own data will carry self-report bias and class imbalance. Finding these now shapes how we collect, clean and label.

2.2 Data description

We use one open dataset now and will collect one dataset of our own.

	Dataset 1: LSApp (open)	Dataset 2: Rota pilot (planned)
Role	Test whether task-to-feed drift exists in real logs	Train and evaluate the drift model
Source	Aliannejadi et al. (2021), released on GitHub	Phase 1: consenting team members; Phase 2: volunteer university students in Türkiye
Collection	Android app (uSearch) logging app events, with consent	Browser extension on YouTube web + Google Takeout watch history + one question per session
Format	TSV: user, session, timestamp, app name, event type	JSON: intention text, video title, channel, timestamp; survey answers (CSV)
Size	3,658,589 rows; 292 users; 76,247 sessions; 87 apps	Phase 1: team members × 2 weeks; Phase 2 target: 40 students × 2 weeks ≈ 1,000 sessions, 15,000 videos
Time span	September 2017 – May 2018; median 5 active days per user	2 weeks per participant
Labels	None	Session: "Did you drift from your goal?" (yes/no); subset of videos: related / unrelated to intention
Access and licence	Public download, released for research; cite the TOIS paper; no separate licence file	Owned by the team; shared only in aggregated, anonymised form

Why these two. LSApp is one of the few open, session-level app-usage logs. Its session IDs let us follow how a session moves from app to app, which is exactly the drift we study, at app level. It has no intentions and no video titles, so it cannot train Rota. The pilot fills that gap: it adds the intention, the content watched and a label from the student.

Pilot instruments. At enrolment, each participant completes the Turkish SAS-SV (Noyan et al., 2015) and a few demographic items. During the study, the extension asks "What are you here for?" when YouTube opens and "Did you drift from your goal?" when the session ends. The SAS-SV score lets us check whether drift relates to problematic use.

2.3 Ethics and data privacy

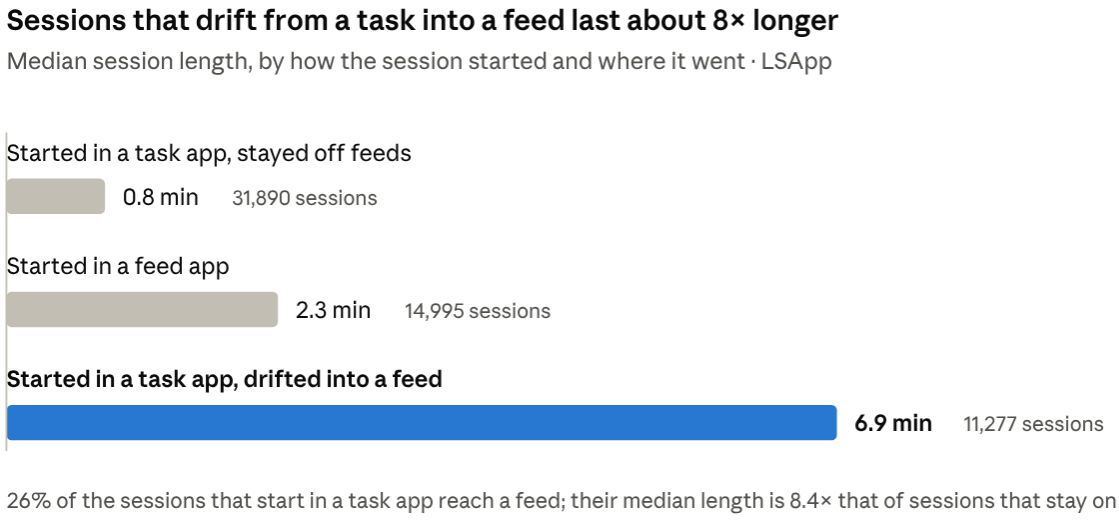
Watch history is sensitive. Video titles can reveal health conditions, religious or political views, and sexual orientation, which KVKK (Law No. 6698) treats as special categories of personal data. We will treat all pilot data as sensitive.

- **Consent.** Explicit, written informed consent before any data is collected, in plain Turkish. The form states what is collected, why, for how long, and that participants can withdraw at any time and have their data deleted. Participants can pause the extension whenever they want.
- **Adults only.** Only students aged 18 or over take part. Minors are excluded, so no parental consent process is needed.
- **Phased pilot.** Phase 1 uses only data from consenting team members who track their own YouTube use, so no outside participants are involved and we do not plan an ethics committee application at this stage; we will confirm this with our programme mentor. Phase 2 opens the pilot to volunteer students only with informed consent, and we will seek ethics committee approval if the programme or a future publication requires it.
- **Data minimisation.** The extension records only what the model needs: intention text, video title, channel and time. No comments, searches, account details or video content. Where possible, titles are turned into embeddings and similarity scores on the participant's device, and only scores and labels leave it.
- **Anonymisation.** Participants get random IDs. The ID-to-name key is stored separately and destroyed after the study.
- **Secure storage.** Encrypted storage with access limited to the team. Data is deleted at the end of the project.
- **Support.** The consent form and the extension include Yeşilay's YEDAM line (115) for anyone concerned about their own use.

LSApp was collected by its authors with participant consent and contains no names. We use it only for aggregate analysis and do not try to re-identify anyone.

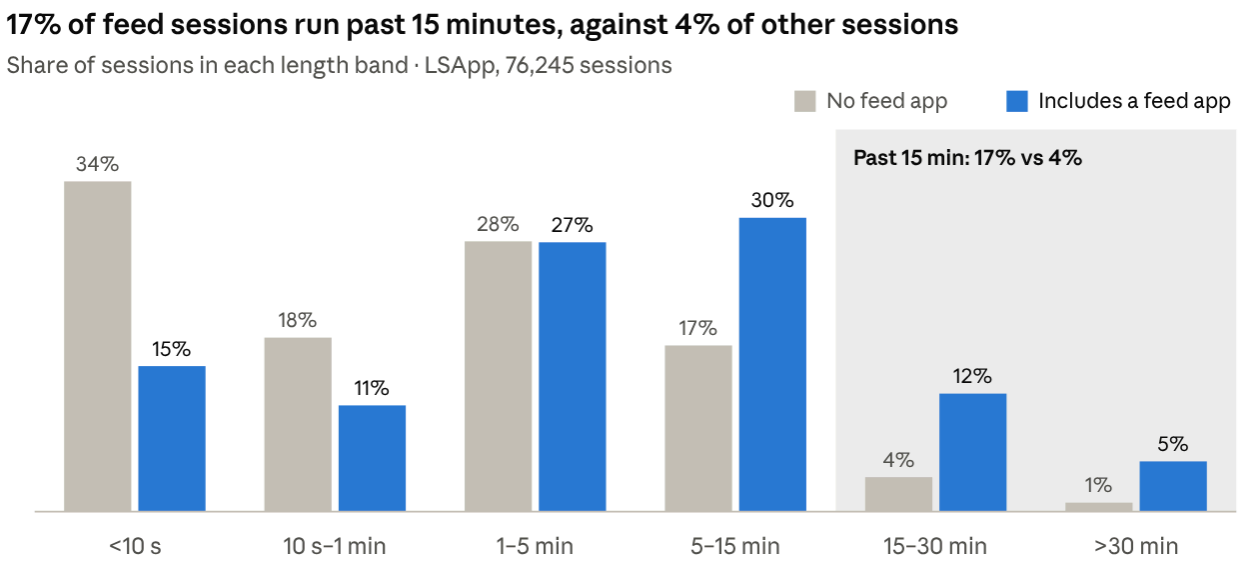
2.4 Data analysis and insights (LSApp)

The pattern Rota targets is clearly visible in real logs: when a session that starts with a task reaches a feed, it becomes about eight times longer. We grouped apps into *feed apps* (Facebook, Instagram, YouTube, Twitter, Reddit, Snapchat, Pinterest, Quora, Flipboard) and *task apps* (messaging, email, phone, search, maps, calendar, camera, settings and similar). A session counts as a feed session if it includes at least one feed app.



LSApp (Aliannejadi et al., 2021) · 43,167 sessions that start in a task app plus 14,995 that start in a feed app · our analysis

Of 43,167 sessions that began in a task app, 26.1% reached a feed. Their median length was 6.9 minutes, against 0.8 minutes for task sessions that stayed off feeds. The effect holds within people: of the 238 users who had both kinds of session, 94.5% had longer drift sessions, with a median ratio of 7.8×. It also holds after dropping zero-length sessions (6.9 vs 1.7 minutes). App logs cannot show *intent*, so this is a proxy for drift, but it shows that drift is common and costly.



LSApp (Aliannejadi et al., 2021) · all 76,245 sessions after cleaning · our analysis

Feed sessions are not only longer on average; they are much more likely to become long. Feed apps appeared in 39.7% of all sessions, and these sessions had a median length of 4.5 minutes against 0.9 minutes for the rest.

Other patterns.

- 89.7% of users opened a feed app at least once; the median user had 21.8 sessions a day.
- Facebook (14,290 sessions), Instagram (8,347) and YouTube (5,745) were the most common feed apps.
- Feeds rarely end a session where it started: 59.1% of sessions that opened in a feed app moved on to other apps.
- The share of sessions that include a feed varied by hour, from 33% to 45%. Timestamps carry no time zone, so we do not draw conclusions about night use from this dataset; our pilot will record local time.

Data quality and how we handle it

Issue	What we found	How we handle it
Duplicate rows	2,185,366 of 3,658,589 rows (59.7%) were exact duplicates	Removed before analysis
Empty sessions	15.3% of sessions last 0 seconds (an app opened and closed in the same second)	Kept in counts, checked separately; results hold without them

Issue	What we found	How we handle it
Broken events	4,485 "Broken" events	Removed; 2 sessions disappeared as a result (76,247 → 76,245)
Unclear time zone	Timestamps have no time zone	No time-of-day conclusions; pilot logs local time
Short coverage	Median user active on 5 days	Results reported per session and per user
Different population	US participants, 2017–2018, before TikTok's rise; not students	Used only to show the pattern exists, not to train Rota
Record count mismatch	README reports 599,635 records; the file holds 3,658,589 rows	Our figures come from the file itself

We expect three further issues in our own pilot. **Class imbalance**: most sessions will not drift, so we will use class weights and report F1 rather than accuracy. **Self-report bias**: students may under-report drift, so we combine their answer with the measured similarity curve and check agreement on a hand-labelled subset. **Missing answers**: some end-of-session questions will be skipped; we drop those sessions from training but keep them for describing usage.

2.5 Conclusion

The open data confirms the problem Rota is built around. About a quarter of task sessions drift into a feed, and when they do they last about eight times longer, a pattern that holds for nearly every user. What app logs cannot provide is the intention and the content watched, so they can show *that* drift happens but not *how far* a session strayed from its goal. Our pilot dataset is designed to fill exactly that gap, with intentions, video titles, a student-provided label and the Turkish SAS-SV, under KVKK-compliant consent and minimisation.

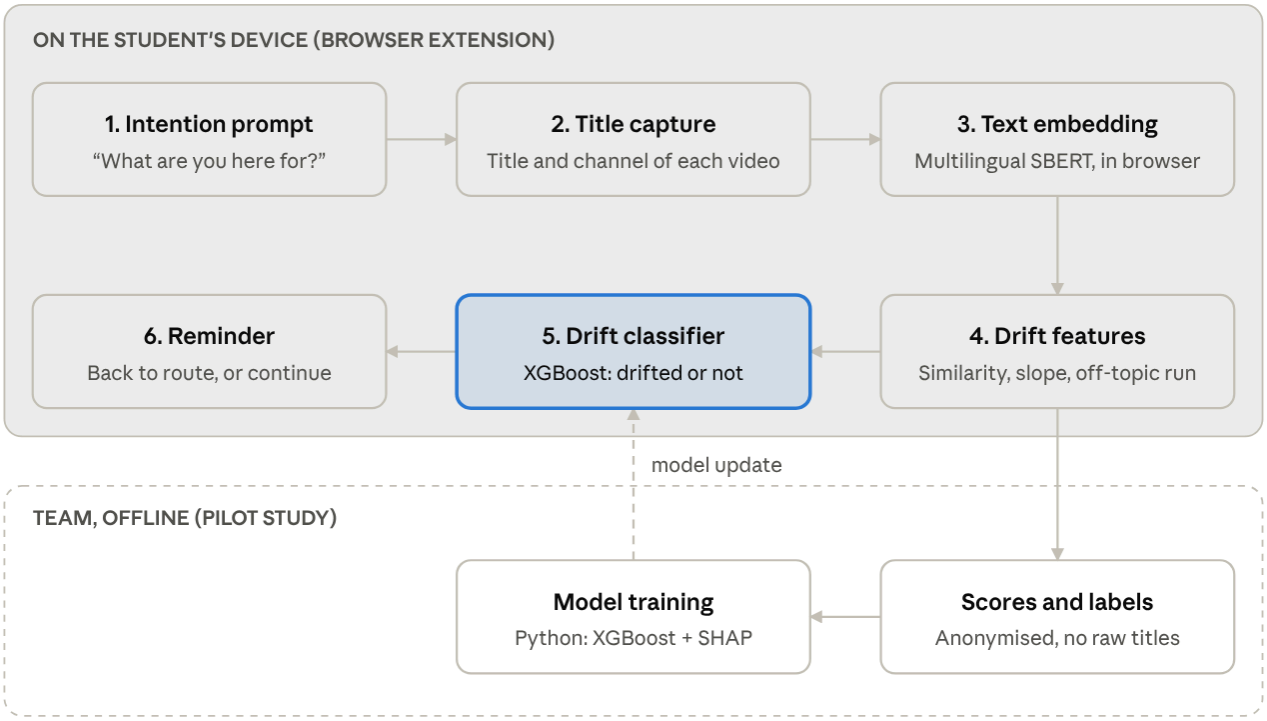
Part 3 — Technology Review

3.1 Introduction

Rota needs four kinds of technology: a way to **capture** the intention and the videos watched, a way to **understand** whether a video matches the intention, a **model** that decides when a session has drifted, and a way to **deliver** the reminder. Each choice

affects accuracy, cost and, above all, privacy. This review compares the options so that we can pick tools that fit a student project and protect our participants.

Raw titles stay on the device; only scores and labels reach the team



Rota system architecture · 6 on-device steps, 2 offline steps

The intention and the video titles are processed inside the browser. Only similarity scores, model features and the student's own label leave the device for training.

3.2 Technology overview and relevance

Capturing the session

- **Browser extension (Chrome, Manifest V3).** A content script runs on YouTube pages, shows the intention prompt and reads each video's title and channel from the page. *Relevance:* it is the only option that gives us the intention and the content in real time, so it is our pilot platform.
- **Google Takeout watch history.** Users can export their YouTube history as JSON (title, channel, time). *Relevance:* a free, user-controlled backup source, useful for cross-checking the extension's logs. It has no intentions and arrives only after the fact.
- **YouTube Data API v3.** Returns video metadata such as category and tags, with a default quota of 10,000 units a day. *Relevance:* can add category and tag context to short or vague titles.

- **Android UsageStatsManager.** Gives app-level foreground events once the user grants the special *Usage access* permission. *Relevance:* needed for a later mobile version, but it reports *which* app was open, not *what* was watched inside it. This is the same limit we saw in LSApp.
- **Apple Screen Time API.** Lets apps set limits and show shields, but keeps detailed usage data inside Apple's privacy layer. *Relevance:* useful for blocking, too closed for drift detection.

Understanding the content

- **Keyword overlap (TF-IDF).** Counts shared words between intention and title. Fast and transparent, but it misses synonyms: "mercimek çorbası" and "lentil soup" share no words.
- **Sentence embeddings (Sentence-BERT).** Map a sentence to a vector so that similar meanings land close together (Reimers & Gurevych, 2019). The multilingual model *paraphrase-multilingual-MiniLM-L12-v2* covers 50 languages, outputs 384-dimensional vectors and is Apache-2.0 licensed. *Relevance:* it matches meaning across Turkish and English titles, which is essential for Turkish students. We will check its Turkish quality on a hand-labelled sample.
- **Large language models (LLMs).** Can judge "is this video related to the goal?" with no training. They are strong at meaning, but every call sends watch history to a third party, costs money and adds delay.
- **Transformers.js / ONNX Runtime Web.** Run small models like MiniLM directly in the browser. *Relevance:* lets us compute embeddings on the device, so raw titles never leave it.

Deciding when a session has drifted

- **Fixed threshold.** Flag a session when similarity stays below a cut-off for N videos. Simple and our first baseline.
- **Logistic regression.** A linear model on session features; a second, interpretable baseline.
- **XGBoost.** Gradient-boosted decision trees that work well on small tabular datasets and handle missing values (Chen & Guestrin, 2016). *Relevance:* about 1,000 labelled sessions suits it well, and the trained model is small enough to run in the browser.
- **LSTM or Transformer over video sequences.** Learn drift directly from the order of videos. Promising, but they need far more labelled sessions than a pilot provides. This is our next step if the data grows.
- **SHAP.** Explains each prediction by showing how much each feature pushed it. *Relevance:* lets us tell a student *why* a reminder appeared, and lets us check the model for bias.

3.3 Comparison and evaluation

Option	Cost	Ease for our team	Performance on our task	Privacy	Explainability	Verdict
Keyword overlap (TF-IDF)	Free	Very easy	Weak: misses synonyms and translation	High: runs locally	High	Baseline only
Multilingual SBERT (MiniLM)	Free, open source	Easy (Python and browser)	Strong for short titles, cross-language	High: runs on device	Medium: a similarity score	**Chosen** for meaning
LLM API	Paid per call	Easy to try	Strongest at nuance	Low: history sent to a provider	Medium: gives reasons, not always reliable	Rejected for the pilot
Fixed threshold	Free	Very easy	Fragile: one cut-off for everyone	High	High	Baseline only
Logistic regression	Free	Easy	Moderate	High	High	Baseline
XGBoost + SHAP	Free	Moderate	Strong on small tabular data	High: small model, runs locally	High with SHAP	**Chosen** for decisions
LSTM / Transformer	Free, compute-heavy	Hard	Potentially strongest with big data	High if on device	Low	Future work
Chrome extension	Free	Moderate	Real-time intention and titles	High: local processing	Open code	**Chosen** for the pilot
Android UsageStatsManager	Free	Moderate	App-level only, no content	Medium	High	Later mobile version

On accessibility for students: every chosen tool is free and open source, and the extension works on any laptop with Chrome. Students on phones are covered later through Android.

3.4 Use cases and examples

Example	Who	What it does	What it shows for Rota
YouTube "Take a break" reminder	YouTube	Pauses the video after a chosen time; on by default for ages 13–17, off for adults	Time-based and opt-in, so most adult students never see it
YouTube Shorts daily limit (2025)	YouTube	Pauses the Shorts feed after a daily limit; users can dismiss it	Platforms now accept feed limits, but still count minutes
Digital wellbeing dashboards and app timers	Android, iOS	Screen time charts, app limits, focus and bedtime modes	Liked, but do not build new habits (Monge Roffarello & De Russis, 2019)
one sec app	Commercial app	Short pause before a chosen app opens	Developer-reported 57% drop in use of the target app (one sec, n.d.)
SwitchTube (research)	University of Washington and partners	YouTube client with a search-first Focus mode and a recommendation-first Explore mode	Users rated Focus higher than Explore on agency and goal alignment (46 users, 3 weeks; Lukoff et al., 2023)
Türkiye Bağımlılıkla Mücadele Eğitim Programı (TBM)	Yeşilay	Prevention training with a technology addiction module	Awareness reaches students, but not at the moment of use
YEDAM	Yeşilay	Free counselling for technology and other addictions; hotline 115	The professional help Rota should refer to

The closest precedent is SwitchTube: like Rota, it treats intention as the key and found that a search-first mode improves agency. Its limit is that the user must choose the mode. Rota adds automatic detection of the moment the session drifts.

3.5 Gaps, risks and research opportunities

Gaps in existing tools.

- Every tool above measures *time* or *app*. None knows what the user came for, so none can tell a purposeful hour from a lost one.
- Most limits are opt-in for adults and easy to dismiss without reflection.
- Language models that could judge relevance mostly run in the cloud, which conflicts with privacy for watch history.
- Mobile platforms expose app-level usage only. Content-level drift detection on phones would need platform support or an accessibility-service workaround that raises its own privacy questions.

Customisation we need. A Turkish-capable similarity model checked on Turkish titles; a feature set built around the intention (similarity, slope, whether the intention was ever reached); an in-browser classifier; and a reminder designed with students rather than for them.

Risks of our own solution, and how we address them.

Risk	Mitigation
Rota becomes another source of compulsive use	No points, streaks, badges or leaderboards. At most one reminder per session and a daily cap. No notifications outside YouTube.
Reminders feel like nagging and get ignored	"Continue on purpose" is always one tap away; the reminder respects that choice and learns from it.
False alarms on legitimate exploration	Students can set an "explore" intention that turns drift detection off for that session, as SwitchTube's Explore mode does.
Privacy of watch history	Embeddings are computed on the device; only scores and labels are shared, under KVKK-compliant consent; data is deleted after the study.
Model bias against some topics or languages	Check accuracy separately for Turkish and English titles and across content categories; use SHAP to inspect what drives predictions.

Risk	Mitigation
Rota is mistaken for treatment	Rota is a self-awareness tool, not a diagnosis. If a student's pattern looks serious (frequent long drift sessions, high SAS-SV score), Rota shows a short message pointing to **YEDAM (115)** and never offers counselling itself.

3.6 Conclusion

The review points to a clear stack. A Chrome extension captures the intention and the titles. A multilingual Sentence-BERT model, running in the browser, measures how far each video is from the intention. An XGBoost classifier, explained with SHAP, decides when a session has drifted. These tools are free, fit a dataset of about 1,000 sessions, explain their decisions and keep raw watch history on the student's device. Existing products count minutes or block apps; this combination lets Rota respond to what actually matters, the gap between what a student meant to do and what they are doing.

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